The University of Texas at Austin
Homework Assignment 9

Predictive Analytics

April 13, 2026

Instructions: Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

Decision Trees.

Problem 9.1. (10 points) Solve Problem **8.4.1** from page 361 from the textbook.

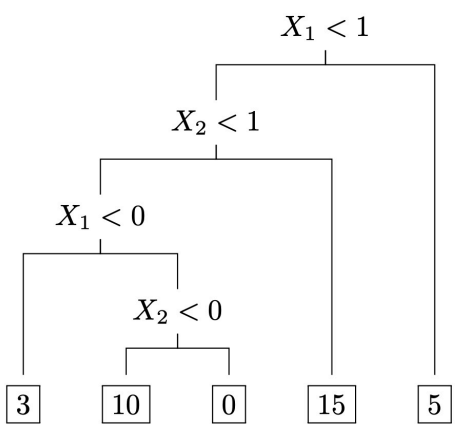
Solution. Solutions will vary.

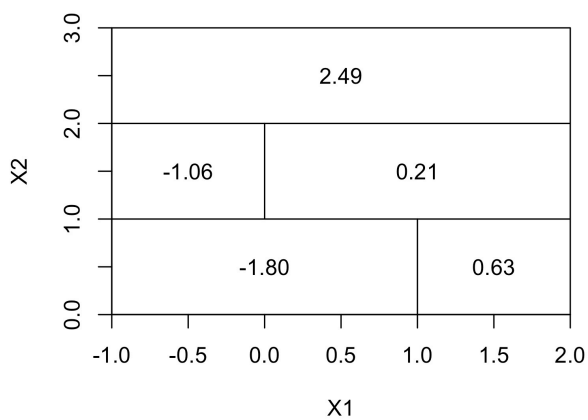
Problem 9.2. (5 points) Draw an example of a partition in the plane that **cannot** possibly correspond to recursive binary splitting.

Solution. Solutions will vary.

Problem 9.3. (10 points) Solve Problem **8.4.4** from page 362 from the textbook.

Solution. The tree and the partition should look something like this (up to various symmetries):





Problem 9.4. (10 points) *Source: An old SRM manual.*
Consider the following observations of (X, Y) with X being the predictor and Y being the response:

$$(0, 8), \quad (1, 5), \quad (3, 8), \quad (6, 6).$$

After one iteration of recursive binary splitting, there are two groups of observations. Find the members of the two groups.

Solution. Remember that - in general - the criterion for choosing the splits is to minimize the residual sum of squares (RSS)

$$\sum_{j=1}^{J} \sum_{i \in R_j} \left(y_i - \hat{y}_{R_j}\right)^2$$

where $\hat{y}_{R_j}$ stands for the average of the response variable in region R_j for $j = 1, ..., J$. However, since this problem is computationally too complex, we resort to f recursive binary splitting. Hence, as there is one predictor only in our current problem, we must make the split along its possible values. Every available split is f binary and partitions the support of X into R and R^c. So, it creates an RSS with this structure

$$\sum_{i \in R} (y_i - \hat{y}_R)^2 + \sum_{i \in R^c} (y_i - \hat{y}_{R^c})^2.$$

In this problem, we can now proceed "by hand" from the lowest to the highest observed value of the predictor.

If $(0, 8)$ is the sole element in R, the mean response for the remaining points is

$$\frac{5+8+6}{3} = \frac{19}{3}.$$

The RSS is

$$\left(5 - \frac{19}{3}\right)^2 + \left(8 - \frac{19}{3}\right)^2 + \left(6 - \frac{19}{3}\right)^2 = \frac{14}{3}.$$

If $(0, 8)$ and $(1, 5)$ form the first region, the mean response in that region is $\frac{13}{2}$. The remaining points $(3, 8)$ and $(6, 6)$ are in the other region and their 7. So, the RSS is

$$\left(8 - \frac{13}{2}\right)^2 + \left(5 - \frac{13}{2}\right)^2 + (8-7)^2 + (6-7)^2 = \frac{13}{2}.$$

If $(0, 8)$, $(1, 5)$, and $(3, 8)$ are in the first region and only $(6, 6)$ remains in the other region, then the average of the first region's values of the response variable is

$$\frac{8+5+8}{3} = 7.$$

So, the RSS equals

$$(8-7)^2 + (5-7)^2 + (8-7)^2 = 6.$$

Overall, the smallest RSS corresponds to the first partition with $(0, 8)$ in its own region, and the remaining points in the other region.

Problem 9.5. (15 points) *Source: Sample MAS-II.*
A data set contains six observations for two predictor variables, X_1 and X_2, and a response variable Y. Here is the table of observations:

X_1	X_2	Y
1	0	1.2
2	1	2.1
3	2	1.5
4	1	3.0
2	2	2.0
1	1	1.6

The following five splits are analyzed:

- $R_1(1,1) = \{X \mid X_1 < 1\}$ and $R_2(1,1) = \{X \mid X_1 \geq 1\}$
- $R_1(1,4) = \{X \mid X_1 < 4\}$ and $R_2(1,4) = \{X \mid X_1 \geq 4\}$
- $R_1(2,0) = \{X \mid X_2 < 0\}$ and $R_2(2,0) = \{X \mid X_2 \geq 0\}$
- $R_1(2,1) = \{X \mid X_2 < 1\}$ and $R_2(2,1) = \{X \mid X_2 \geq 1\}$
- $R_1(2,2) = \{X \mid X_2 < 2\}$ and $R_2(2,1) = \{X \mid X_2 \geq 2\}$

Determine which split is chosen first.

Solution. First note that f I. and f III. do not constitute a meaningful partition of the predictor space (since all observations end up in R_2). Now, let's focus on the other proposed binary splits individually

In f II., the only point in R_2 is $(4, 1, 3.0)$. All the remaining points are in R_1. The mean of the observations in R_1 is

$$\frac{1.2 + 2.1 + 1.5 + 2.0 + 1.6}{5} = 1.68.$$

So, the RSS equals

$$(1.2 - 1.68)^2 + (2.1 - 1.68)^2 + (1.5 - 1.68)^2 + (2.0 - 1.68)^2 + (1.6 - 1.68)^2 = 0.548.$$

In f IV., $(1, 0, 1.2)$ is the only point in R_1. The mean of the response in the remaining observations is

$$\frac{2.1 + 1.5 + 3.0 + 2.0 + 1.6}{5} = 2.04$$

So, the total RSS equals

$$(2.1 - 2.04)^2 + (1.5 - 2.04)^2 + (3.0 - 2.04)^2 + (2.0 - 2.04)^2 + (1.6 - 2.04)^2 = 1.412.$$

In f V., $(3, 2, 1.5)$ and $(2, 2, 2.0)$ are placed into R_2 while the remaining points end up in R_1. The mean of the response values in R_2 is 1.75. So, the contribution to the RSS from R_2 equals

$$(1.5 - 1.75)^2 + (2.0 - 1.75)^2 = 0.125$$

The mean of the response values in R_1 is

$$\frac{1.2 + 2.1 + 3.0 + 1.6}{4} = 1.975.$$

So, the contribution to the RSS from R_1 equals

$$(1.2 - 1.975)^2 + (2.1 - 1.975)^2 + (3.0 - 1.975)^2 + (1.6 - 1.975)^2 = 1.8075$$

The total RSS is 1.9325.

Since its RSS is the smallest of the ones proposed, **II.** is our final answer.